$U \leq k^n$: k-vector subspace

$\begin{array}{c} \text{\rotatebox{90}{$\cong$}} \\ k^m \end{array}$

Proposition 2 $\quad k^n/U \simeq k^{n-m}$

Observation 1

$$k^m \xrightarrow{\simeq} U \leq k^n$$

$\rightsquigarrow \quad k^m \longrightarrow k^n$: k-linear transform. (injective) with image U

Suggests: $k^m \xrightarrow{T} k^n$: k-linear transform.

Try to understand $k^n/\operatorname{im} T$.

Observation (2)

$$\left\{ \begin{array}{c} \text{Linear transformations} \\ T: k^m \longrightarrow k^n \end{array} \right\}$$

$$\begin{array}{c} T \\ \updownarrow \\ (T(e_1), \dots, T(e_m)) \end{array} \Big\updownarrow$$

$$\left\{ (v_1, \dots, v_m) \in \underbrace{k^n \times \dots \times k^n}_{m\text{-times}} \right\}$$

$$\begin{array}{c} (v_1, \dots, v_m) \\ \updownarrow \\ [v_1 \ \dots \ v_m] \end{array} \Big\updownarrow$$ View $v_i \in k^n$ as an $(n \times 1)$-column vector

$$\left\{ A \in M_{n \times m}(k) \right\}$$

$$(a_1, \dots, a_n) \longleftrightarrow \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

i.e. Every linear transformation

$$T : k^m \longrightarrow k^n$$

is represented by the $(n \times m)$-matrix

$$A_T = [T(e_1)\ T(e_2) \cdots T(e_m)]$$

Observation ③

$$\begin{tikzcd} k^m \arrow[r, "T"] & k^n \arrow[d, "\psi"', "\simeq"] \\ k^m \arrow[u, "\varphi"', "\simeq"] \arrow[r, "\psi \circ T \circ \varphi"'] & k^n \end{tikzcd}$$

Then (a) $k^n / \operatorname{im} T \simeq k^n / \operatorname{im}(\psi \circ T \circ \varphi)$

(b) $\underset{\underset{M_{m \times m}(k)}{\cap}}{A_\varphi}, \underset{\underset{M_{n \times n}(k)}{\cap}}{A_\psi}$ are invertible matrices i.e. $\exists$ matrix inverses for A_φ, A_ψ

$$\Leftrightarrow \quad A_\varphi \in GL_m(k) = \left\{ A \in M_{m\times m}(k) : \exists B \text{ s.t. } AB = I_m \right\}$$

$$A_\psi \in GL_n(k)$$

(c) $A_{\psi \circ T \circ \varphi} = A_\psi A_T A_\varphi$

Rmk: In (b) & (c), we are using the following:

$$\begin{tikzcd} k^n \arrow[r, "T"] \arrow[dr, "S\circ T"'] & k^m \arrow[d, "S"] \\ & k^r \end{tikzcd} \quad \rightsquigarrow \quad A_{S\circ T} = \underbrace{A_S A_T}_{\text{matrix product}}$$

Upshot: To compute $k^n/\operatorname{im} T$ up to isomorphism, we can change A_T by row & column operations

Proposition:

$A \in M_{n \times m}(k)$

After row & column operations, A can be put in the following form:

$$\begin{bmatrix} 1 & & 0 & 0 \\ & 1 & & \\ 0 & & 1 & 0 \quad 0 \\ & & \ddots & \\ 0 & 0 & 0 & 0 \quad 0 \end{bmatrix}$$

i.e. (i) All non-zero entries are along the diagonal & equal 1.

(ii) If a diagonal entry is 0 then all subsequent diagonal entries are also 0.

Observation 4

$$A_T = \left.\underbrace{\begin{bmatrix} \overbrace{\begin{matrix} 1 & & \\ & 1 & \\ & & \ddots\ 1 \end{matrix}}^{r \text{ columns}} \Big\} r \text{ rows} & 0 \\ 0 & \begin{matrix} 0 & & \\ & \ddots & \\ & & 0 \end{matrix} \end{bmatrix}}_{m\text{-columns}}\right\} n\text{-rows}$$

Then, $\operatorname{im} T = k^r \times \overbrace{\{0\} \times \{0\} \times \cdots \times \{0\}}^{n-r} \leq k^n$

$$k^n / \operatorname{im} T = \frac{k^r \times k^{n-r}}{k^r \times \{0\}^{n-r}}$$

$$\simeq k^r / k^r \times k^{n-r} / \{0\}^{n-r}$$

$$\simeq k^{n-r}.$$

Putting all of this together finishes the proof of Proposition 2

Corollary 1

$$k^n \simeq k^m \implies n = m$$

Pf:

$$T : \underline{k^m} \xrightarrow{\simeq} k^n$$

$$k^n / \operatorname{im} T \overset{\text{Prop 2}}{\simeq} k^{n-m}$$

$$\|$$

$$k^n / k^n \simeq \{0\}$$

$$\implies n = m$$

Corollary 2: W : f.g. k-vector space

Then $W \simeq k^d$ for a unique $d \geq 0$.

Pf: $\exists\, U \leq k^n$ s.t. $k^n / U \simeq W$

Proposition (1) k^m

Now Proposition (2) says

$$k^{n-m} \simeq k^n/U \simeq W$$

$$\Rightarrow W \simeq k^d \quad \text{where } d = n-m$$

(Uniqueness)

$$W \simeq k^d \ \& \ W \simeq k^e$$

$$\Rightarrow k^d \simeq k^e \xRightarrow{\text{Corollary}} d = e$$

We can put A into

reduced row echelon form

(RREF)

i.e. a matrix with the following properties:

(i) All non-zero entries are along the diagonal